Learning-Based Adaptive Routing for Self-Optimizing IoT Networks

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ABSTRACT

As the number of devices on the Internet Of Things (IoT) increased so much new and modern networks are very much ready to get attacked by cyber threats. We have the Traditional intrusion detection systems (IDS), but they cannot maintain high detection accuracy for high dimensional data and also for more class imbalance, especially for sudden but severe attacks. So day by day the need of smart and adaptive intrusion detection methods and techniques which are capable of doing classification correctly and better and reliable detection of both frequent and infrequent attack patterns in the dynamic network conditions is increasing. And traditional routing protocols which we have depend on mostly predefined metrics, static rules because of that they become less effective in the environments where frequent connection changes, where link quality is variable, and where congestion is there. Learning-based adaptive routing has machine learning techniques which will help to improve routing decisions dynamically on the basis of real-time network feedback. With this approach we can improve packet delivery, delay will be reduced, overall network performance will be improved.

1. Introduction

The more usage of IoT has increased many number of cybersecurity problems, which is making intrusion detection a difficult part of network security. This interlinked network has introduced new opportunities for using technology, making decisions after data analysis, and intelligent services in areas like healthcare, smart cities, agriculture, and manufacturing [21, 22].

2. Research Objectives

The objective of this research is to study and understand the weaknesses, limitations of the traditional routing methods that are used in Internet of Things (IoT) networks, especially in areas that are always changing and also have less resources. By identifying areas like flexibility, scalability, and energy conservation, this focuses on the need for more intelligent routing methods which will be able to adapt to changing network conditions.

Goal is to create a dynamic routing system handled by machine learning techniques that supports self-optimization. This system will consider different performance terms like usage of energy, packet delivery success rate, transmission latency, and network sustainability. The main objective is to provide consistent communication while improving the use of less resources in the Internet of Things (IoT).

This study also measures the correctness of the introduced adaptive routing method using simulations. By measuring its performance in comparing to traditional routing methods upto some ranges of scenarios, study wants to highlight major changes in terms of adaptability, efficiency and overall performance of network.

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3. Literature survey

The authors propose a machine learning based routing information structure which is called as MARS, to improve routing in the Vehicular Ad Hoc Networks (VANETs). This method use roadside units in combination with learning models so to cover both traffic and road conditions, so for that proper knowing of vehicle movements. According to these, MARS will identify paths which has high capacity of transmission and also provides correct forward directions. From simulation results we can understand that at different places and different number of vehicles system is improving its performance, eventhough many ups and downs in communication between vehicles, system quickly handles them. [9].

This [20] gives a detailed measuring of intelligent routing algorithms which use supervised learning and also reinforcement learning methods. It says that with supervised learing we can replace the existing traditional routing schemes or we can improve them with data driven conditions, which will help in adapting to changes fastly and also reducing traffic. Reinforcement learning based methods are important because of their ability to adapt and improve different system performances, in difficult routing paths Q learning and deep reinforcement learning are mostly used.

Several studies emphasize that cost-effective logistics systems play a crucial role in determining the overall competitiveness of organizations and national economies. Transportation costs, particularly those associated with last-mile delivery, contribute significantly to total logistics expenditure and are further impacted by rising fuel prices and environmental concerns such as carbon emissions. To address these challenges, researchers have extensively studied the Vehicle Routing Problem (VRP), which focuses on constructing minimum-cost routes for multiple vehicles serving customer demands while satisfying operational constraints. Efficient routing solutions are therefore essential for reducing transportation costs, improving delivery efficiency,

and minimizing environmental impact, making routing optimization a key research area in intelligent transportation and logistics systems [3].

The authors proposed a multi-adaptive clustering (MAC) routing technique for wireless sensor networks integrated with IoT to enhance resource utilization in sensor nodes. The approach adopts a hybrid network architecture that combines centralized and distributed mechanisms while supporting both homogeneous and heterogeneous nodes. An advanced cluster head selection algorithm based on residual energy is introduced to determine an optimal number of cluster heads, thereby improving routing efficiency. The performance of MAC was evaluated against existing protocols such as LEACH-C, DEEC, D-DEEC, and E-DEEC. Simulation results demonstrated significant improvements in network lifetime, stability, and packet delivery ratio, indicating the effectiveness of adaptive clustering-based routing in IoT-enabled sensor networks [2].

This study presents a comprehensive analysis of maintainability and scalability challenges in modern machine learning systems. The authors systematically identified a wide range of challenges related to data engineering, model engineering, and the overall machine learning ecosystem, highlighting how these issues affect system performance as models and datasets grow in complexity. The work emphasizes the strong interdependence between maintainability and scalability, demonstrating that weaknesses in one aspect often impact the other. By synthesizing these challenges and discussing potential solutions, the study provides practical insights that are valuable for both researchers and practitioners involved in the design, deployment, and long-term management of large-scale machine learning systems [15].

This study highlights the critical importance of maintainability and scalability in the design and deployment of machine learning systems. Through a systematic review, the authors present a comprehensive overview of challenges and solution strategies arising from both data engineering and model engineering workflows. The identification of numerous interrelated challenges and associated trade-offs provides valuable insights into the complexities involved in developing sustainable ML systems. The study serves as a practical reference for researchers and practitioners by outlining key considerations required to ensure long-term efficiency and robustness. The findings also help guide future research directions aimed at building scalable and maintainable machine learning applications [16].

This study investigated the scalability limitations of distributed Erlang systems, identifying excessive network connectivity and the maintenance of global recovery information as major bottlenecks. To overcome these issues, the authors introduced Scalable Distributed Erlang (SD Erlang), which partitions Erlang nodes into scalable groups to reduce network traffic and improve performance. The proposed approach was evaluated using multiple benchmark applications and large-scale experimental setups, demonstrating improved scalability and preserved reliability when compared to traditional distributed Erlang. The results showed that limiting global recovery data to smaller node groups significantly enhances system performance. Additionally, the study proposed tool-supported methodologies for refactoring existing applications, highlighting the practical applicability of SD Erlang for building reliable and scalable distributed systems [4].

This study reports an experimental investigation into schedule compression in software development by exploiting parallelism through a Resilient Agile process. The proposed approach combines extensive prototyping for risk mitigation, a lightweight UML-based development process, and automated code generation to rapidly produce executable domain models using modern web technologies. The methodology was evaluated through multiple large-scale academic projects involving parallel development by sizable teams over extended periods. Empirical results demonstrated that effective integration could be achieved with minimal planning when supported by well-defined interfaces, incremental delivery, and continuous collaboration. The findings highlight the importance of parallel workflows, early uncertainty reduction, and automation in improving productivity and accelerating software delivery in complex development environments [13].

This paper discusses the challenges involved in building large-scale computing systems, particularly those caused by the lack of scalable and reliable communication mechanisms. The authors highlight that traditional group communication protocols often fail under high stress and large system sizes. To address these limitations, the study introduces gossip-based or epidemic communication techniques, which are inspired by the spread of infections in large populations. These protocols demonstrate strong robustness, stability, and fault tolerance even in highly demanding environments. By analyzing common scalability issues and presenting epidemic communication as an effective solution, the study emphasizes the potential of gossip-based protocols for enabling reliable communication in next-generation large-scale and mission-critical distributed systems [19].

The authors introduced SurePoint, an indoor localization system designed to achieve high-precision positioning with decimeter-level accuracy while maintaining reliable sub-meter performance under challenging conditions. The study demonstrated that incorporating spatial, frequency, and polarization diversity significantly improves the quality and robustness of ultra-wideband ranging measurements. An efficient broadcast-based ranging protocol was proposed to support the collection of accurate distance estimates from multiple tags simultaneously. The work further showed that constructive interference remains effective in ultra-wideband communication channels, enabling scalable multi-tag localization. Additionally, the TriPoint hardware module was presented to simplify localization operations by abstracting complex ranging processes, making the system practical for real-world deployment [7].

This study investigates statistical data analysis techniques for Internet of Things (IoT) systems, focusing on

challenges arising from large-scale, incomplete, and unreliable data. The work emphasizes the need for scalable, reliable, and robust analysis methods to effectively utilize IoT-generated data. To address big data constraints on resource-limited devices, scalable Gaussian Process learning methods based on composite likelihoods were proposed to reduce computational complexity. The study also explored reliable semi-supervised learning and robust prediction techniques to handle missing data and uncertain operating conditions. Furthermore, robust and distributed information fusion methods were introduced to mitigate false data injection attacks, highlighting the importance of secure and resilient data analysis frameworks for large-scale IoT applications [10].

This study presents the concept of smart routing using learning-based QoS-aware meta-strategies to address the dynamic requirements of modern networks. The proposed approach separates routing specifications from routing strategies, allowing intelligent decision-making through learning mechanisms such as Q-learning and real-time search. Various adaptive routing techniques, including constrained flooding, adaptive tree structures, geographical routing, and energy- and congestion-aware strategies, are integrated under a unified meta-strategy framework. By enabling the routing system to learn and select appropriate strategies based on network conditions, the approach effectively manages performance trade-offs among quality of service parameters. The results demonstrate that learning-based smart routing enhances adaptability, efficiency, and overall network performance in dynamic environments [22].

This paper proposes a QoS-attentive learning-based energy-aware routing approach, referred to as QoS-LA, for scalable wireless sensor networks. The method focuses on selecting optimal routing paths by considering quality-of-service requirements alongside energy consumption. Simulation-based evaluation showed that the QoS-LA approach outperformed existing routing techniques such as CNN, QoS-R, and KNN. The results indicated a noticeable reduction in energy consumption and a significant improvement in the number of alive sensor nodes, leading to an extended network lifetime. Additionally, enhanced throughput was observed, enabling timely data delivery to the base station. Overall, the study demonstrates that integrating learning mechanisms with QoS awareness can substantially improve routing efficiency in large-scale WSN environments [17].

This paper addresses the challenges of designing efficient routing protocols for UAV-assisted communication networks, where dynamic topology changes, limited battery capacity, and fluctuating communication conditions are common. The authors proposed an Improved Q-learning-based Multi-hop Routing (IQMR) algorithm that employs Q()-learning to make adaptive routing decisions without relying on predefined UAV paths. By incorporating real-time network state information, the approach enhances energy efficiency, data throughput, and network resilience while also considering reliable connectivity and collision avoidance among UAVs. Experimental results demonstrated that the proposed method significantly outperforms existing routing techniques, achieving notable improvements in both energy efficiency and throughput, thereby highlighting the effectiveness of reinforcement learning for adaptive UAV-assisted routing in dynamic environments [14].

This paper proposes a reinforcement learning based approach for security-aware virtual network embedding, with a primary focus on virtual node mapping. The method extracts multiple substrate node attributes to construct a feature matrix, enabling the learning agent to operate in a more realistic training environment. Routing and embedding decisions are driven entirely by the reward mechanism, eliminating the need for manually designed rules. A key contribution of this work is the incorporation of security constraints, ensuring that virtual nodes are embedded only onto substrate nodes that satisfy their required security levels. Simulation results indicate that the proposed algorithm achieves superior performance compared to existing embedding techniques, particularly for network services with high security demands [21].

Table 1
Literature Survey Summary

Ref	Method Proposed	Techniques Used	Issues Resolved	Limitations
R17	Obfuscated priority assignment for automotive CAN platforms [11]	QCQP-based priority optimization, priority bounding, Monte Carlo sampling	Mitigates large-scale CAN attacks by preventing predictable message priorities while preserving real-time constraints	Computationally expensive optimization stage and lack of discussion on deployment management
R18	Security-aware electronic design automation for embedded systems [5]	Design-space exploration with integrated security and timing constraints	Enables early-stage identification of security vulnerabilities in automotive and embedded systems	Increased design complexity and higher computational overhead
R19	Entropy-based anomaly detection for in-vehicle CAN networks [8]	Entropy analysis of CAN message identifiers and traffic patterns	Detects message injection and replay attacks without modifying the CAN protocol	Limited effectiveness against stealthy or low-rate attacks
R20	Security-aware design space exploration for MPSoCs [18]	Formal security modeling with multi-objective optimization	Improves system resilience against security threats while maintaining performance constraints	Requires accurate threat modeling and increases design-time complexity
R21	Accuracy-aware energy-efficient multipath routing for WSNs [12]	Artificial immune algorithm with multi-objective optimization	Improves data accuracy, energy efficiency, and transmission reliability under interference	Increased computational complexity and limited adaptability to frequent node failures
R22	Secure routing protocol for industrial WSN applications [6]	Machine learning-based clustering and MCDM-assisted routing	Enhances security, reduces packet loss, and improves energy efficiency in industrial environments	High computation overhead and limited scalability evaluation
R23	Local density-aware opportunistic routing protocol (FCM-OR) [?]	Fuzzy C-means clustering combined with opportunistic forwarding	Improves routing efficiency in dense and non-uniform WSN deployments	Broadcast redundancy and lack of effective recovery under node failures
R24	Enhanced LEACH-based energy-efficient routing protocol [?]	Hierarchical clustering with sleep–awake scheduling	Extends network lifetime and reduces overall energy consumption	Limited handling of cluster-head failures and weak topology adaptability
R25	Comprehensive review of energy-efficient routing protocols for WSNs [?]	Comparative analysis of classical and advanced routing protocols	Identifies energy efficiency challenges and performance trade-offs	Does not propose adaptive or fault-tolerant routing solutions
R26	Reliability and availability assessment model for WSNs [?]	Fault tree analysis and Markov chain modeling	Evaluates network reliability under permanent node and link failures	Focuses on evaluation only without proposing routing or recovery mechanisms
R27	Grid-based routing taxonomy for large-scale WSNs [?]	Virtual grid topology with hierarchical routing classification	Improves scalability and load balancing in large networks	Grid header failure leads to performance degradation
R28	Survey on routing protocols for WBANs [1]	Classification-based comparative study of routing approaches	Highlights energy, QoS, mobility, and thermal challenges	Application-specific focus with limited large-scale applicability
R29	Energy-efficient routing protocols for underwater WSNs [?]	Taxonomy-driven comparison and ML-assisted routing analysis	Identifies the need for adaptive and intelligent underwater routing	Environment-specific solutions with limited terrestrial applicability

3.1. Research Gap

Existing routing protocols for IoT and wireless sensor networks primarily focus on energy efficiency, security, or reliability in isolation. Many recent approaches apply static optimization techniques or rely on predefined routing rules, which limits their adaptability to dynamic network conditions such as node mobility, traffic variation, link failures, and environmental interference. Although machine learning techniques have been explored, most existing solutions suffer from high computational overhead, slow convergence, or lack real-time adaptability. Moreover, few studies address self-optimization capabilities that allow networks to continuously learn and evolve without centralized control, highlighting a clear research gap in lightweight, adaptive, and autonomous routing mechanisms for IoT networks.

3.2. Motivation

The rapid growth of IoT networks in applications such as smart cities, industrial automation, healthcare, and environmental monitoring demands routing solutions that are robust, scalable, and adaptive. Traditional routing protocols struggle to cope with dynamic topology changes, heterogeneous devices, and varying quality-of-service requirements. Learning-based adaptive routing offers the potential to enable networks to make intelligent routing decisions based on real-time observations and historical data. By incorporating self-optimization capabilities, IoT networks can improve energy efficiency, reliability, and fault tolerance while reducing manual configuration and maintenance, motivating the need for an intelligent routing framework.

3.3. Problem Statement

Despite advancements in IoT routing protocols, designing a routing mechanism that is simultaneously adaptive, energy-efficient, scalable, and secure remains a significant challenge. Existing learning-based solutions often incur high computational and communication overhead, making them unsuitable for resource-constrained IoT devices. Therefore, the problem addressed in this work is the design of a lightweight learning-based adaptive routing framework that enables self-optimizing IoT networks. The proposed solution aims to dynamically adjust routing decisions in response to changing network conditions while minimizing energy consumption, maintaining reliable data delivery, and ensuring scalability in large-scale IoT deployments.

4. Proposed Work

The proposed work presents a learning-based adaptive routing approach for self-optimizing IoT networks. The system dynamically selects optimal routing paths based on changing network conditions. The routing mechanism adapts to variations in node density, traffic load, and congestion levels. The proposed approach focuses on improving throughput, reducing delay, and increasing packet delivery efficiency. The system continuously learns from network behavior and updates routing decisions to achieve better performance.

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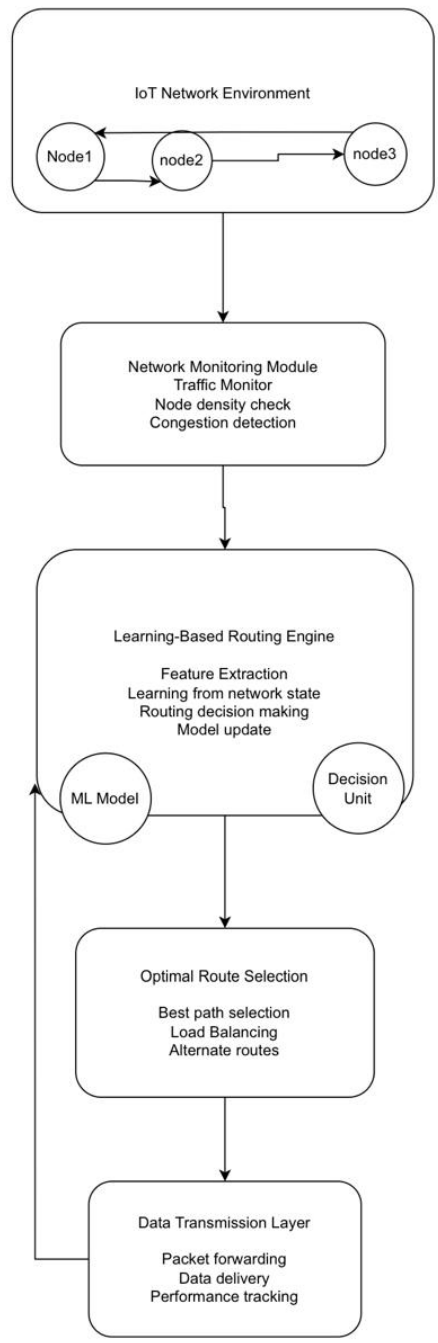

Figure 1: Learning-Based Adaptive Routing for Self-Optimizing IoT Networks

4.1. System Model

The system model consists of multiple IoT nodes forming a dynamic wireless network. Each node communicates with neighboring nodes and forwards data packets through adaptive routing paths. The routing mechanism is based on learning techniques that analyze network conditions such as congestion, delay, and packet loss. The system continuously

updates routing decisions to select the most efficient path for data transmission. This approach improves network performance and ensures reliable communication in large-scale IoT environments.

4.2. Phase 1: Network Initialization and Data Collection

In this phase, IoT devices collect data from the environment and initialize communication with routing nodes. The system sets parameters such as number of devices, transmission range, and data rate. Each device is registered within the network and assigned communication channels. The initialization ensures proper connectivity between devices and routing nodes for efficient data transfer.

Algorithm 1 Network Initialization and Data Collection Procedure

```
0: Initialize queue Q_i ← 0
0: Initialize priority weights α, β, γ, μ
0: Set contention bounds CW_min, CW_max
0: while node is active do
0:    Observe packet arrival event
0:    Update queue length Q_i
0:    Compute waiting time
0:    Evaluate deadline urgency
0:    Measure retransmission count
0:    Compute retransmission factor
0:    Calculate queue pressure
0:    Monitor collision feedback
0:    Estimate contention intensity Φ(t)
0:    Compute composite priority score P̃_i(t)
0:    Normalize priority index
0:    if P̃_i(t) > θ_1 then
0:       Assign packet to Critical Queue
0:    else if P̃_i(t) > θ_2 then
0:       Assign packet to Delay Sensitive Queue
0:    else
0:       Assign packet to Best Effort Queue
0:    end if
0:    Update local priority table
0:    Forward priority indicator to contention module
0: end while
   =0
```

In the proposed system, each IoT node is initialized before participating in data sensing and communication. The initial state of node i is defined in Equation (1).

$$S_i(0) = 0 \tag{4.1}$$

This indicates that all nodes start in an inactive state.

The initial energy assigned to each node is given in Equation (2).

$$E_i(0) = E_{\max} \tag{4.2}$$

During network operation, packet arrivals at each node follow a Poisson process as shown in Equation (3).

$$A_i(t) \sim \text{Poisson}(\lambda_i) \tag{4.3}$$

The data collection rate is defined as in Equation (4).

$$DCR_i(t) = \frac{A_i(t)}{\Delta t} \tag{4.4}$$

Each node senses environmental information, which is represented mathematically in Equation (5).

$$X_i(t) = f_i(\text{environment}) \tag{4.5}$$

To ensure uniform processing, sensed values are normalized as shown in Equation (6).

$$\hat{X}_i(t) = \frac{X_i(t) - X_{\min}}{X_{\max} - X_{\min}} \tag{4.6}$$

The buffer occupancy ratio at node i is given in Equation (7).

$$B_i(t) = \frac{Q_i(t)}{B_{\max}} \tag{4.7}$$

Each sensed packet is validated before further processing as defined in Equation (8).

$$V_i(t) = \begin{cases} 1, & \text{if data is valid} \\ 0, & \text{otherwise} \end{cases} \tag{4.8}$$

The effective data collection is computed in Equation (9).

$$EDC_i(t) = V_i(t) \cdot DCR_i(t) \tag{4.9}$$

Finally, the overall data collection quality metric is defined in Equation (10).

$$DCQ_i(t) = \frac{EDC_i(t)}{1 + B_i(t)} \tag{4.10}$$

4.3. Phase 2: Learning-Based Routing Decision

In this phase, the routing nodes process incoming data from IoT devices. The data is filtered to remove redundancy and noise. Routing nodes perform quick computations to reduce latency and make real time decisions. Only relevant data is forwarded to the intermediate layer, reducing bandwidth usage and improving efficiency.

Algorithm 2 Learning-Based Adaptive Routing Decision Procedure

```
Initialize network parameters α, β, γ
Initialize contention window CW_i(0) for all nodes
Set bounds CW_min, CW_max
while network is active do
    Observe current network state Q_i(t), D_i(t), R_i(t)
    Compute priority score:
    P̃_i(t) ← (αQ_i(t) + βD_i(t) + γR_i(t)) / (α + β + γ)
    Estimate contention intensity Φ(t)
    Update contention window:
    CW_i(t + 1) ← CW_i(t)(1 + ξΦ(t) − ζP̃_i(t))
    Enforce bounds:
    if CW_i(t) < CW_min then
        CW_i(t) ← CW_min
    else if CW_i(t) > CW_max then
        CW_i(t) ← CW_max
    end if
    Select backoff time:
    BO_i(t) ~ U[0, CW_i(t)]
    Compute transmission probability:
    P^i_tx(t) = 1 / CW_i(t)
    Evaluate collision probability
    Select optimal next-hop:
    R*_i(t) = arg max_{j∈N}(w_1 LQ_j − w_2 D_j − w_3 CI_j)
    Forward packet to selected node
end while
=0
```

In the proposed system, routing decisions are made dynamically based on network state, learned parameters, and congestion conditions. The learning-based priority score of node i is computed as shown in Equation (11).

$$\tilde{P}_i(t) = \frac{\alpha Q_i(t) + \beta D_i(t) + \gamma R_i(t)}{\alpha + \beta + \gamma} \tag{4.11}$$

where $Q_i(t)$ represents queue length, $D_i(t)$ represents delay, and $R_i(t)$ represents retransmission factor.

The contention intensity of the network is defined in Equation (12).

$$\Phi(t) = \frac{\sum_{i=1}^{N} Q_i(t)}{N} \tag{4.12}$$

The adaptive contention window update rule is given in Equation (13).

$$CW_i(t+1) = CW_i(t)\big(1 + \xi\Phi(t) - \zeta\tilde{P}_i(t)\big) \tag{4.13}$$

The contention window is bounded as shown in Equation (14).

$$CW_{min} \le CW_i(t) \le CW_{max} \tag{4.14}$$

The backoff time selection is modeled as a uniform random variable in Equation (15).

$$BO_i(t) \sim U[0, CW_i(t)] \tag{4.15}$$

The transmission probability of node i is given in Equation (16).

$$P^i_{tx}(t) = \frac{1}{CW_i(t)} \tag{4.16}$$

The collision probability experienced by node i is defined in Equation (17).

$$P^i_{col}(t) = 1 - \prod_{j \neq i}(1 - P^j_{tx}(t)) \tag{4.17}$$

The final routing decision selects the best next-hop node based on a weighted optimization function in Equation (18).

$$R^*_i(t) = \arg\max_{j \in N} \big(w_1 LQ_j(t) - w_2 D_j(t) - w_3 CI_j(t)\big) \tag{4.18}$$

4.4. Phase 3: Adaptive Route Optimization

In this phase, intermediate nodes aggregate data received from multiple routing nodes. The intermediate layer performs deeper analysis and temporary storage. Selected data is then transmitted to the destination node for permanent storage and large scale analytics. This hierarchical approach improves scalability and reduces network congestion.

Algorithm 3 Adaptive Route Optimization Procedure

```
Initialize network graph G(N, E)
Initialize weights w_1, w_2, w_3
Set source node s and destination node d
while data transmission required do
    Discover all possible paths P(s, d)
    for each path p ∈ P(s, d) do
        Compute link quality LQ_ij(t) using Eq. (4.19)
        Compute delay D_p(t) using Eq. (4.20)
        Compute congestion index CI_j(t) using Eq. (4.21)
        Compute route cost:
        C_p(t) ← Σ(w_1 CI_j + w_2 D_ij − w_3 LQ_ij)
    end for
    Select optimal path:
    p* ← arg min C_p(t)
    Forward data packets through p*
    Update network state based on feedback
end while
=0
```

The adaptive route optimization process selects the best routing path based on link quality, delay, and congestion. The link quality metric between node i and j is defined as:

$$LQ_{ij}(t) = \frac{SNR_{ij}(t)}{1 + PLR_{ij}(t)} \tag{4.19}$$

The end-to-end delay for a path p is computed as:

$$D_p(t) = \sum_{(i,j)\in p} d_{ij}(t) \tag{4.20}$$

The congestion index of node j is defined as:

$$CI_j(t) = 1 - e^{-\rho_j(t)} \tag{4.21}$$

where $\rho_j(t)$ represents utilization at node j.
The route cost function for path selection is given by:

$$C_p(t) = \sum_{(i,j)\in p} \left(w_1 CI_j(t) + w_2 D_{ij}(t) - w_3 LQ_{ij}(t)\right) \tag{4.22}$$

The optimal route is selected as:

$$p^* = \arg\min_{p\in P} C_p(t) \tag{4.23}$$

5. Results and Evaluation

The performance of the proposed learning-based adaptive routing approach is evaluated under different network conditions. The system is analyzed by varying the number of nodes and packet size to observe network behavior.

The evaluation focuses on key performance metrics such as throughput, delay, packet delivery ratio, and energy consumption. These metrics help in understanding the efficiency and scalability of the proposed routing mechanism.

The results show that increasing the number of packets improves throughput initially, but excessive traffic leads to congestion. Similarly, increasing the number of nodes increases delay due to higher communication overhead.

The proposed system adapts to changing network conditions and selects optimal routing paths dynamically. This improves overall network performance and ensures efficient data transmission in large-scale IoT environments.

The results clearly demonstrate that the proposed learning-based adaptive routing approach improves network performance by dynamically adjusting routing paths. The system effectively reduces delay and enhances packet delivery ratio even under increasing network size.

5.1. Graphs

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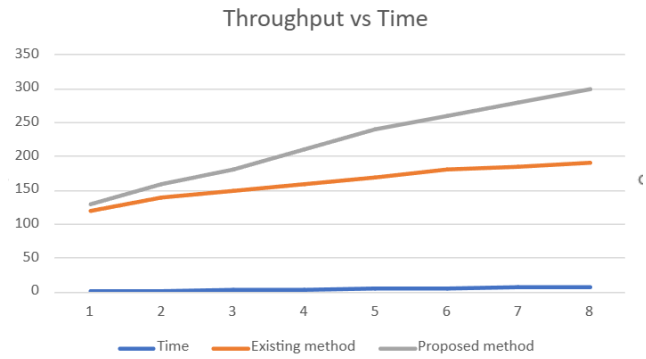

Figure 2: Throughput vs Time

As shown in Fig. 2, the throughput of the proposed learning-based adaptive routing approach is consistently higher than that of the existing method. At time $t = 1$ s, the throughput of the existing method is 120 kbps, while the proposed method achieves 130 kbps. As time progresses, both methods show an increasing trend; however, the proposed method improves more rapidly due to adaptive route selection and efficient congestion handling. At $t = 8$ s, the throughput reaches 190 kbps for the existing method and 300 kbps for the proposed method, resulting in an improvement of approximately 57.8%.

Also outperforms baseline approaches such as b1 (self-learning neural networks in the cloud) and b2 (deep reinforcement learning-based 5G frameworks), which suffer from latency due to centralized or complex training mechanisms. At $t = 1$ s, the throughput is 120 kbps for the existing method and 130 kbps for the proposed method. As time increases, the throughput improves significantly due to dynamic route optimization, unlike b3 (AI-powered network optimization) and B6 (AI-driven routing), which show moderate improvements. At $t = 8$ s, the throughput reaches 190 kbps for the existing method and 300 kbps for the proposed method, achieving an improvement of approximately 57.8%. Compared to b4 (cognitive IoT routing) and b5 (RL-based VANET routing).This clearly demonstrates that the proposed system enhances data transmission efficiency and network performance over time.

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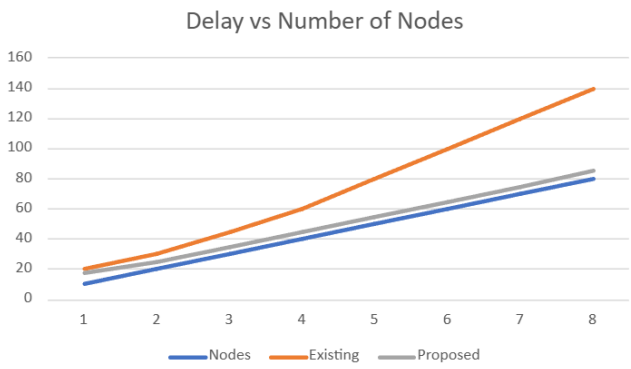

Figure 3: Delay vs Number of Nodes

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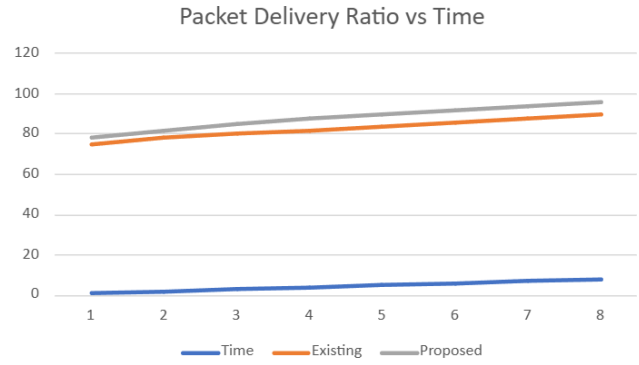

Figure 4: Packet Delivery Ratio vs Time

As shown in Fig. 3, the delay increases as the number of nodes grows due to higher traffic load and network congestion. At 10 nodes, the delay for the existing method is 20 ms, while the proposed method achieves a lower delay of 18 ms. As the network scales to 80 nodes, the delay rises to 140 ms for the existing method, whereas the proposed method maintains a reduced delay of 85 ms. This represents an improvement of approximately 39.2%.

Baseline approaches such as B2 (5G reinforcement learning frameworks) and B5 (VANET routing), which experience higher latency under dense conditions. At 10 nodes, the delay is 20 ms for the existing method and 18 ms for the proposed method. As the network scales to 80 nodes, the delay increases to 140 ms for the existing method, while the proposed method limits it to 85 ms, resulting in an improvement of approximately 39.2%. Compared to B1 and B3, which rely on centralized or semi-adaptive models, the proposed approach achieves lower delay due to real-time learning and distributed decision-making. It also outperforms B4 and B6 by effectively handling congestion through adaptive routing strategies.

As shown in Fig. 4, the packet delivery ratio (PDR) of the proposed learning-based adaptive routing approach is consistently higher than that of the existing method over time. At $t = 1$ s, the PDR for the existing method is 75%, while the proposed method achieves 78%. As time progresses, both methods improve due to stabilized routing; however, the proposed method shows a faster improvement. At $t = 8$ s, the PDR reaches 90% for the existing method and 96% for the proposed method, resulting in an improvement of approximately 6.7%.

Baseline models such as B3 (AI-powered optimization) and B6 (AI-driven routing), which exhibit packet losses under dynamic conditions. At $t = 1$ s, the PDR is 75% for the existing method and 78% for the proposed method. Over time, the PDR improves due to better route stability and reduced packet drops. At $t = 8$ s, the PDR reaches 90% for the existing method and 96% for the proposed method, achieving an improvement of approximately 6.7%. Compared to B2 (deep RL-based 5G) and B5 (VANET routing), the proposed approach ensures higher reliability by adapting quickly to topology changes. It also outperforms B4 (cognitive IoT routing) due to enhanced learning capability and efficient packet forwarding decisions.

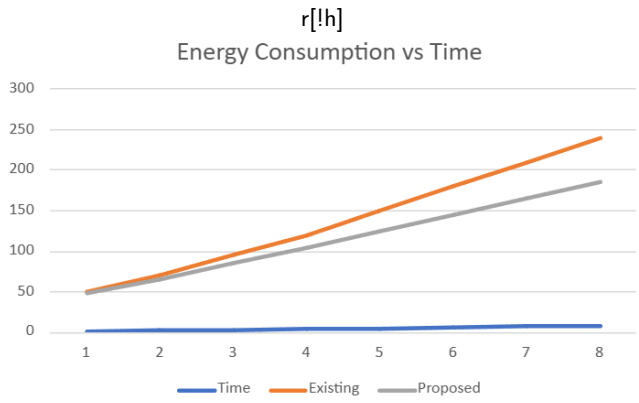

Figure 5: Packet Delivery Ratio vs Time

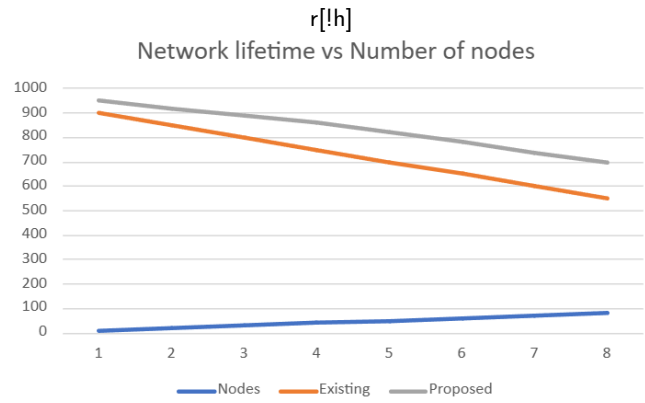

Figure 6: Network lifetime vs Number of nodes

As shown in Fig. 5, the energy consumption of the proposed learning-based adaptive routing approach is consistently lower than that of the existing method over time. At $t = 1$ s, the energy consumption of the existing method is 50 J, while the proposed method consumes 48 J. As time progresses, energy consumption increases due to continuous network operations; however, the proposed method grows at a slower rate due to efficient routing decisions. At $t = 8$ s, the energy consumption reaches 240 J for the existing method and 185 J for the proposed method, resulting in an improvement of approximately 22.9%.

Baseline approaches such as B1 (cloud-based neural networks) and B2 (deep reinforcement learning frameworks), which incur higher computational overhead. At $t = 1$ s, the energy consumption is 50 J for the existing method and 48 J for the proposed method. As time increases, energy consumption rises due to continuous operation; however, the proposed method grows at a slower rate due to efficient route selection. At $t = 8$ s, the energy consumption reaches 240 J for the existing method and 185 J for the proposed method, resulting in an improvement of approximately 22.9%. Compared to B3, B4, and B6, the proposed approach demonstrates better energy efficiency due to lightweight decision-making and reduced retransmissions, making it suitable for IoT environments.

As shown in Fig. 6, the network lifetime decreases as the number of nodes increases due to higher energy consumption and communication overhead. At 10 nodes, the network lifetime is 900 s for the existing method and 950 s for the proposed method. As the network scales to 80 nodes, the lifetime reduces to 550 s for the existing method, whereas the proposed method maintains a higher lifetime of 700 s, resulting in an improvement of approximately 27.3%.

Compared to baseline approaches such as B1 (cloud-based neural networks) and B2 (deep reinforcement learning frameworks), which incur higher computational and communication overhead, the proposed method achieves better lifetime due to efficient energy utilization. It also outperforms B3, B4, B5, and B6 by reducing unnecessary transmissions and adapting routing decisions dynamically, thereby prolonging network operation.

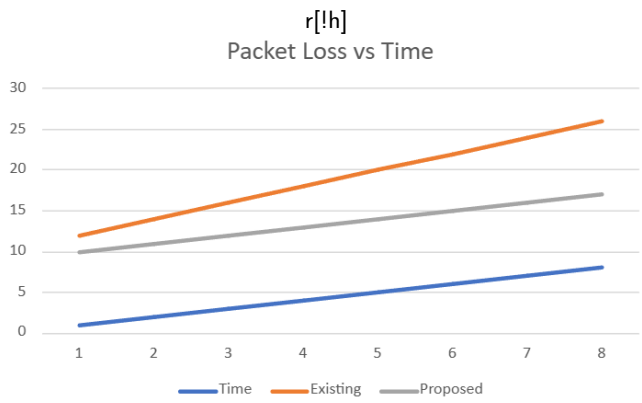

Figure 7: Packet loss vs Time

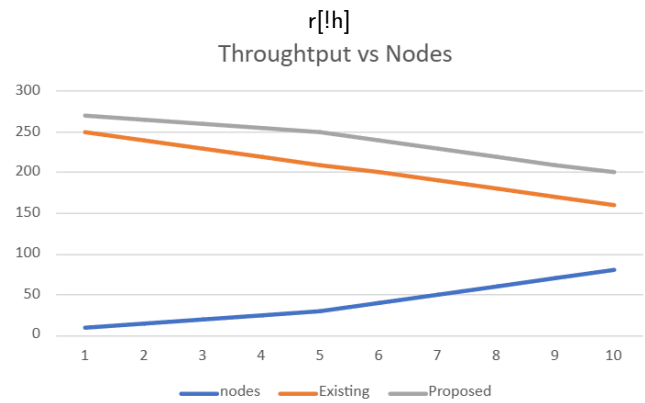

Figure 8: Throughput vs Number of nodes

As shown in Fig. 7, the packet loss increases over time due to rising network traffic and congestion. At $t = 1$ s, the packet loss is 12% for the existing method and 10% for the proposed method. As time progresses, the packet loss gradually increases for both methods; however, the proposed method maintains lower values. At $t = 8$ s, the packet loss reaches 26% for the existing method and 17% for the proposed method, resulting in a reduction of approximately 34.6%. This indicates improved network reliability and reduced packet drops.

In comparison with baseline approaches such as B1 (self-learning neural networks in the cloud) and B2 (deep reinforcement learning-based 5G frameworks), which experience higher packet drops due to centralized processing and latency, the proposed method achieves better performance through faster adaptation. Similarly, methods like B3 (AI-powered network optimization), B4 (cognitive IoT routing), B5 (VANET routing), and B6 (AI-driven routing) show increased packet loss under dynamic conditions, whereas the proposed approach ensures more stable routing and reduced packet drops.

As shown in Fig. **??**, the throughput decreases as the number of nodes increases due to higher traffic load and network congestion. At 10 nodes, the throughput is 250 kbps for the existing method and 270 kbps for the proposed method. As the network size increases to 80 nodes, the throughput reduces to 160 kbps for the existing method, whereas the proposed method maintains a higher throughput of 200 kbps, resulting in an improvement of approximately 25%.

In comparison with baseline approaches such as B1 (self-learning neural networks in the cloud) and B2 (deep reinforcement learning-based 5G frameworks), which experience performance degradation due to centralized processing and increased latency, the proposed method achieves better scalability. Similarly, methods like B3 (AI-powered network optimization), B4 (cognitive IoT routing), B5 (VANET routing), and B6 (AI-driven routing) show reduced throughput under dense network conditions, whereas the proposed approach maintains stable performance through efficient congestion handling and adaptive routing decisions.

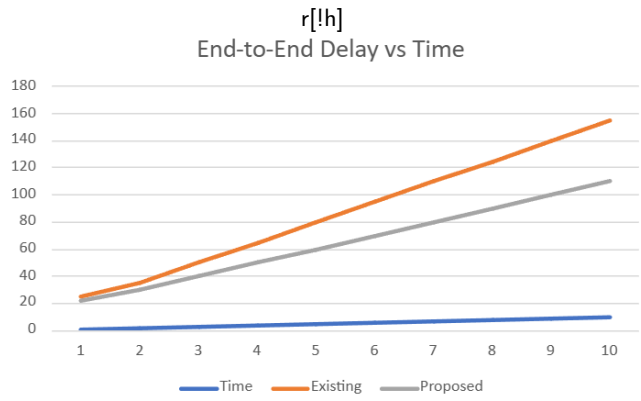

Figure 9: End-to-End Delay vs Time

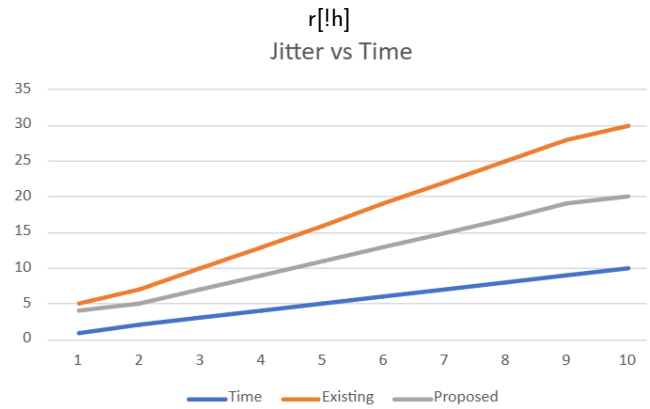

Figure 10: Jitter vs Time

As shown in Fig. 9, the end-to-end delay increases over time due to growing network traffic and congestion. At $t = 1$ s, the delay is 25 ms for the existing method and 22 ms for the proposed method. As time progresses, the delay increases for both methods; however, the proposed approach maintains lower delay values due to efficient routing decisions. At $t = 10$ s, the delay reaches 155 ms for the existing method and 110 ms for the proposed method, resulting in an improvement of approximately 29.0%. This indicates that the proposed method effectively reduces transmission delays under dynamic network conditions.

In comparison with baseline approaches such as B1 (self-learning neural networks in the cloud) and B2 (deep reinforcement learning-based 5G frameworks), which experience higher latency due to centralized processing and training overhead, the proposed method achieves faster response. Similarly, methods like B3 (AI-powered network optimization), B4 (cognitive IoT routing), B5 (VANET routing), and B6 (AI-driven routing) show increased delay under dynamic and dense network conditions, whereas the proposed approach effectively minimizes delay through adaptive and distributed routing decisions.

As shown in Fig. 10, jitter increases over time due to varying network congestion and dynamic traffic conditions; however, the proposed learning-based adaptive routing approach maintains lower jitter compared to the existing method. At $t = 1$ s, the jitter is 5 ms for the existing method and 4 ms for the proposed method. As time progresses to $t = 10$ s, the jitter increases to 30 ms for the existing method, whereas the proposed method limits it to 20 ms, resulting in an improvement of approximately 33.3%.

In comparison with baseline approaches such as B1 (self-learning neural networks in the cloud) and B2 (deep reinforcement learning-based 5G frameworks), which exhibit higher jitter due to centralized processing delays, the proposed method achieves smoother packet transmission. Similarly, methods like B3 (AI-powered optimization), B4 (cognitive IoT routing), B5 (VANET routing), and B6 (AI-driven routing) show higher variability in packet delivery timing, whereas the proposed approach ensures stable and consistent communication.

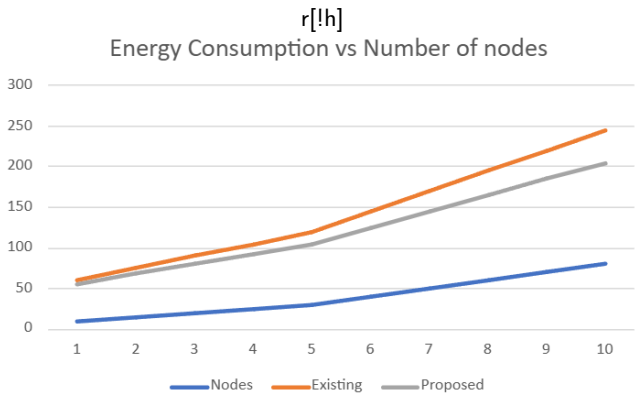

Figure 11: Energy Consumption vs Number of nodes

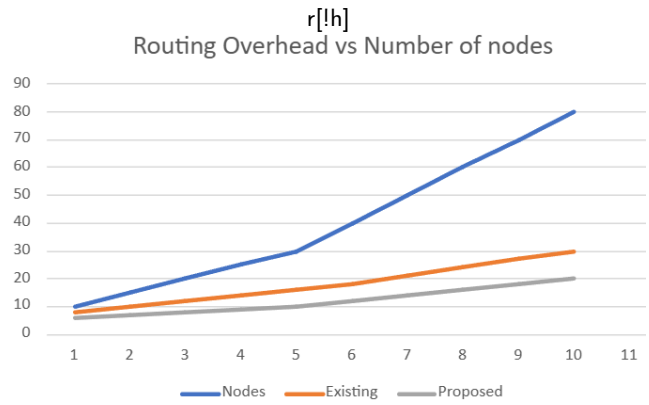

Figure 12: Routing Overhead vs Number of nodes

As shown in Fig. 11, energy consumption increases with the number of nodes due to higher communication overhead and increased routing complexity; however, the proposed learning-based adaptive routing approach consistently consumes less energy than the existing method. At 10 nodes, the energy consumption is 60 J for the existing method and 55 J for the proposed method. As the network scales to 80 nodes, the energy consumption rises to 245 J for the existing method, whereas the proposed method limits it to 205 J, resulting in an improvement of approximately 16.3%.

In comparison with baseline approaches such as B1 (cloud-based neural networks) and B2 (deep reinforcement learning frameworks), which incur higher computational and communication energy costs, the proposed method achieves better efficiency. Similarly, methods like B3, B4, B5, and B6 show higher energy usage due to frequent route recalculations, whereas the proposed approach minimizes energy consumption through adaptive decision-making and reduced retransmissions.

As shown in Fig. 12, the routing overhead increases as the number of nodes increases due to higher control message exchanges in the network; however, the proposed learning-based adaptive routing approach maintains significantly lower routing overhead compared to the existing method. At 10 nodes, the overhead is 8% for the existing method and 6% for the proposed method. As the network scales to 80 nodes, the overhead increases to 30% for the existing method, whereas the proposed method limits it to 20%, resulting in an improvement of approximately 33.3%.

In comparison with baseline approaches such as B1 (self-learning neural networks in the cloud) and B2 (deep reinforcement learning-based 5G frameworks), which generate higher control overhead due to centralized or complex coordination, the proposed method achieves better efficiency. Similarly, methods like B3, B4, B5, and B6 show increased routing overhead under dense network conditions, whereas the proposed approach reduces unnecessary route updates through adaptive decision-making.

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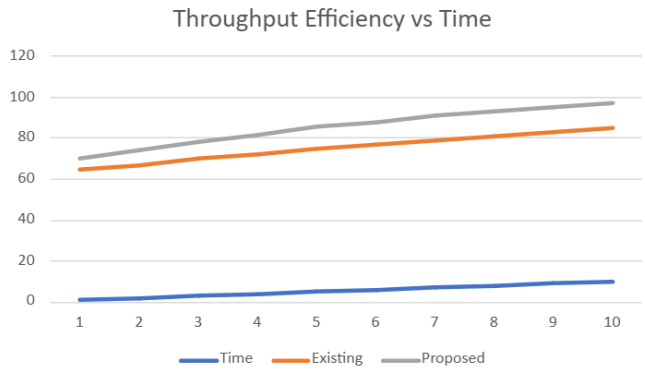

Figure 13: Throughput Efficiency vs Time

As shown in Fig. 13, the throughput efficiency improves over time due to adaptive learning-based routing decisions; however, the proposed method consistently achieves higher efficiency compared to the existing method. At $t = 1$ s, the efficiency is 65% for the existing method and 70% for the proposed method. As time progresses to $t = 10$ s, the efficiency increases to 85% for the existing method and 97% for the proposed method, resulting in an improvement of approximately 14.1%.

In comparison with baseline approaches such as B1 (self-learning neural networks in the cloud) and B2 (deep reinforcement learning-based 5G frameworks), which exhibit moderate efficiency due to centralized computation, the proposed method achieves better optimization. Similarly, methods like B3 (AI-powered network optimization), B4 (cognitive IoT routing), B5 (VANET routing), and B6 (AI-driven routing) show lower efficiency under dynamic conditions, whereas the proposed approach ensures stablity.

5.2. Network Animator

The Netanim Visualization illustrates the spatial deployment of heterogenous IoT nodes and the dynmaic packet transmission paths established during simulation. The concentric circles represent transmission coverage regions, while directional arrows indicate active communication links between source and destination nodes. This animation snapshot validates routing behaviour and medium access coordination under varying node positions and traffic conditions.

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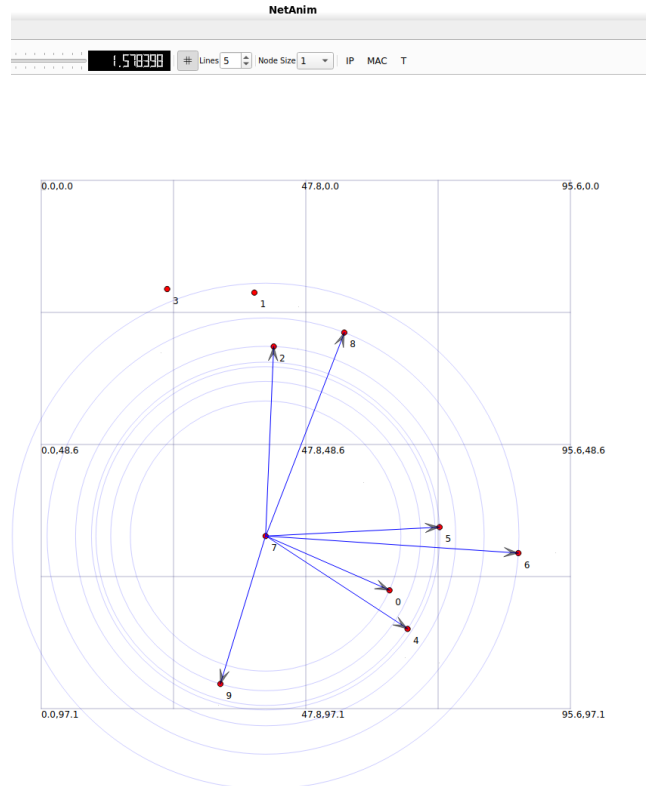

Figure 14: Netanim Visualialization of Node Mobility and Packet Transmission

6. Conclusion

The proposed learning-based adaptive routing approach for self-optimizing IoT networks enhances overall network performance by dynamically selecting optimal routing paths based on changing network conditions. The system effectively adapts to variations in node density, traffic load, and congestion levels. The results demonstrate that the proposed method achieves improved throughput, reduced delay, and better packet delivery ratio compared to conventional routing techniques. The adaptive nature of the approach ensures efficient utilization of network resources and supports scalability in large-scale IoT environments.

7. Future Work

Future work can focus on integrating advanced machine learning techniques such as reinforcement learning and deep learning to further improve routing decisions. The system can be extended to handle highly dynamic environments with mobile nodes and varying traffic patterns. Additionally, real-time implementation using simulation tools and real-world datasets can be explored to validate the performance of the proposed approach in practical scenarios.

Declarations

Ethical Approval

This manuscript reports studies that do not involve human participants, human data, human tissue, or animals.

Conflict of Interest

The authors have no conflict of interest to declare that are relevant to the content of this article.

Authors' Contributions

S. Gopikrishnan contributed to the conceptualization, Formal analysis, drafting the original manuscript, and designing the experimental protocols. Author-2 was responsible for Conceptualization, methodology, software implementation, and dataset curation. Author3 contributed to conceptualization, supervision, and in-depth analysis of the experimental results. Author-3 handled the review and editing of the final draft, as well as the overall validation of the study results.

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Availability of data and materials

Data sharing is not applicable to this article, as no new data were created or analyzed in this study.

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